

Project	Demo Project	Title	Beam Calculation Report
Element ID: B1		Revision: R0	Date: 2026-06-19
Calculation	Simply Supported Beam v1		

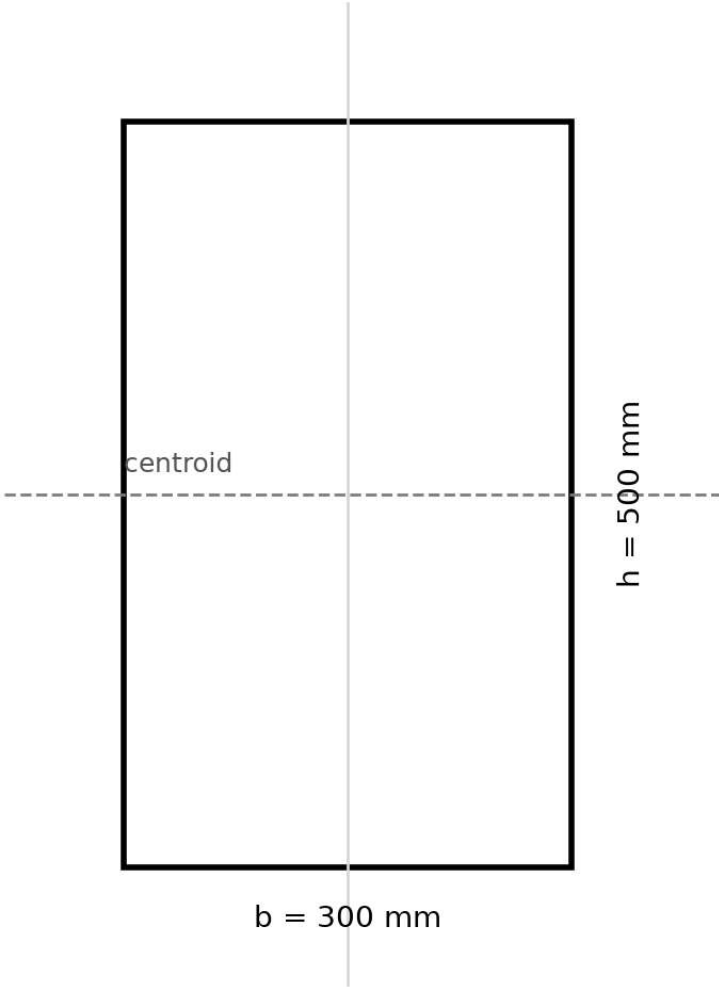
Input

Inputs

Parameter	Symbol	Value
Span	L	5 m
Uniformly distributed load	w	20 kN/m
Section width	b	300 mm
Section height	h	500 mm
Characteristic compressive strength of concrete	f_{ck}	30 MPa

Section drawing

Rectangular cross-section



Gross rectangular beam section.

Project	Demo Project	Title	Beam Calculation Report
Element ID: B1		Revision: R0	Date: 2026-06-19
Calculation	Simply Supported Beam v1		

Final values

Final Values

Parameter	Symbol	Value
Maximum bending moment	M_{Ed}	62.5 kNm
Characteristic compressive strength of concrete	f_{ck}	30.0 MPa
Maximum shear force	V_{Ed}	50.0 kN
Section area	A	150000.0 mm^2
Second moment of area	I	3125000000.0 mm^4
Top fibre bending stress	σ_{top}	-5.0 N/mm^2
Bottom fibre bending stress	σ_{bottom}	5.0 N/mm^2

1. Maximum bending moment

$$M_{Ed} = \frac{wL^2}{8}$$
$$M_{Ed} = \frac{20.000 \times (5.000)^2}{8}$$
$$M_{Ed} = 62.500 \text{ kNm}$$

Reference: Basic beam formula: simply supported beam under UDL

1.1 Characteristic compressive strength of concrete

$$f_{ck} = f_{ck}$$
$$f_{ck} = 30.000$$
$$f_{ck} = 30.000 \text{ MPa}$$

Reference: Characteristic compressive strength of concrete

2. Maximum shear force

$$V_{Ed} = \frac{wL}{2}$$
$$V_{Ed} = \frac{20.000 \times (5.000)}{2}$$
$$V_{Ed} = 50.000 \text{ kN}$$

Reference: Basic beam formula: simply supported beam under UDL

Project	Demo Project	Title	Beam Calculation Report
Element ID: B1		Revision: R0	Date: 2026-06-19
Calculation	Simply Supported Beam v1		

3. Section area

$A = bh$
 $A = 300 \times (500)$
 $A = 150000 \text{ mm}^2$

Reference: Rectangular section area

4. Second moment of area

$I = \frac{bh^3}{12}$
 $I = \frac{300 \times (500)^3}{12}$
 $I = 3125000000 \text{ mm}^4$

Reference: Rectangular section about centroidal strong axis

5. Top fibre bending stress

$\sigma_{top} = \frac{-M_{Ed} 10^6 \frac{h}{2}}{I}$
 $\sigma_{top} = \frac{-62.500 \times 10^6 \times \frac{500.000}{2}}{3125000000.000}$
 $\sigma_{top} = -5.000 \text{ N/mm}^2$

Reference: Elastic bending stress: sigma = My/I

6. Bottom fibre bending stress

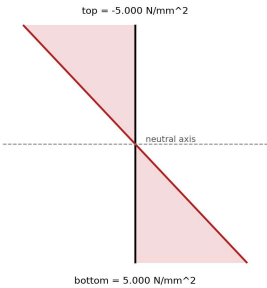
$\sigma_{bottom} = \frac{M_{Ed} 10^6 \frac{h}{2}}{I}$
 $\sigma_{bottom} = \frac{62.500 \times 10^6 \times \frac{500.000}{2}}{3125000000.000}$
 $\sigma_{bottom} = 5.000 \text{ N/mm}^2$

Reference: Elastic bending stress: sigma = My/I

Project	Demo Project	Title	Beam Calculation Report
Element ID: B1		Revision: R0	Date: 2026-06-19
Calculation	Simply Supported Beam v1		

7. Bending stress distribution

Bending stress distribution



Linear elastic stress distribution for sagging bending.

8. Stress distribution through section depth

Stress distribution through section depth 2

Position	y from centroid (mm)	Stress (N/mm^2)
Top fibre	-250.000	-5.000
Point 2	-194.444	-3.889
Point 3	-138.889	-2.778
Point 4	-83.333	-1.667
Point 5	-27.778	-0.556
Point 6	27.778	0.556
Point 7	83.333	1.667
Point 8	138.889	2.778
Point 9	194.444	3.889
Bottom fibre	250.000	5.000